10.1 Creating Box Plots

Lesson Introduction

Benchmarks	MA.6.DP.1.3 Given a box plot within a real-world context, determine the minimum, the lower quartile, the median, the upper quartile, and the maximum. Use this summary of the data to describe the spread and distribution of the data. MA.6.DP.1.5 Create box plots and histograms to represent sets of numerical data within real-world contexts.	Learning Targets	• I can find key values given a box plot within a real-world context. • I can create a box plot to represent data.

	Building On	Working Toward
Benchmark Coherence	MA.5.DP.1.1 Collect and represent numerical data, including fractional and decimal values, using tables, line graphs, or line plots.	MA.7.DP.1.5 Given a real-world numerical or categorical data set, choose and create an appropriate graphical representation.

Essential Questions	• What information does the graph give you about the data? • What is an advantage of displaying the data in a graph rather than a list of values? • Why would you choose a box plot graph for this data?
Lesson Narrative	In this lesson, students use box plots to represent data sets. This lesson begins with students reviewing vocabulary related to data sets and extends this knowledge to representing data using box plots. Students apply what they know to create box plots to represent given data sets using the five-number summary, and the lesson culminates with a self-assessment of the lesson's learning.

Component Summary	10.1.1 Warm-Up (~5 min.)	Calculate the median and range for a group of data *(SE p. 203)*	10.1.4 Guided Practice (5 min.)	Create a box plot with a set of data points with assistance *(SE p. 206)*
	10.1.2 Refresher (5-10 min.) (optional)	Label a blank box plot and review vocabulary *(SE p. 204)*	10.1.5 Try It (5-10 min.)	Create a box plot with a set of data points *(SE p. 207)*
	10.1.3 Guided Instruction (15 min.)	Complete the five-number summary for a two data sets *(SE p. 205)*	10.1.6 Your Turn! (10 min.)	With a partner, discuss the five-number summary and complete a box plot *(SE p. 208)*
	10.1.7 Wrap-Up (~5 min.)	Reflection – Rank your understanding of today's assignment *(SE p. 209)*		

	Glossary	Materials	Additional Preparation
Resources	<u>Key Terms</u>: • Five-number summary <u>Additional Terms:</u> • Box plot • Interquartile range • Median • Range • Minimum • Maximum • First quartile • Third quartile	• (Optional) Foldable/interactive notebook	• N/A

<table>
<tr><td rowspan="4">Teacher Tips</td><td>Common Misconceptions</td><td>Things to Consider</td></tr>
<tr><td>Students might confuse the percentages contained in a given interval of the boxplot.
Students may mistakenly label the median as the mean.
Students may misread the graph, especially if comparing two graphs.
Students may forget to order data before finding the five-number summary.</td><td>Stress the percentage breakdown of the quartiles in the graph, discussing the percentage of data values above, below, and in between values.
Have students find the five-number summary before creating the box plot.
Discuss that while the box plot displays the overall shape of the distribution, individual data values and unusual features such as gaps, clusters, and outliers cannot be determined from the box plot.
If the data contains larger values, the x-axis does not have to start at zero. Use the range of the data to determine an appropriate scale.
The IQR represents the middle 50 percent of the data.</td></tr>
<tr><td>Literacy and Language Routines</td><td>Mathematical Thinking and Reasoning (MTR) Standard</td></tr>
<tr><td>Notice and Wonder
10.1.3 Guided Instruction provides an opportunity to engage students in a Notice and Wonder routine to position students to consider how to represent data sets with an even and odd numbers of data points using box plots.</td><td>MA.K12.MTR.2.1: Multiple Representations
Students will be working to represent box plots. Clarifications within this standard emphasize connections between concept and representation. Utilize the progression of the lesson to highlight the process by which students are moving from finding the data to representing it in a box plot.</td></tr>
<tr><td>Differentiate Instruction</td><td colspan="2">10.1.4 Guided Practice - Consider differentiating this component by ability. Use informal data gathered earlier in the lesson to group students. Allow students who demonstrate understanding to work independently, students who may benefit from discussion to work in pairs and pull a small group of students who are struggling to guide them through this component verbally.

The 10.1.7 Wrap-Up requires a written explanation as part of a self-assessment. For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.</td></tr>
<tr><td colspan="3">Lesson Notes:</td></tr>
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10.1.1 Warm-Up: Sleep

Component Overview: *Students find the median and range of a data set, two measures used to help construct box plots later in this lesson.*

10.1.1 Warm-Up: Sleep

Ten sixth-grade students were asked how much sleep, in hours, they usually get on a school night. Their responses are shown.

$$6, 7, 6.5, 7.5, 8, 8.5, 7, 6.5, 5, 7$$

1. Calculate the median number of hours of sleep for the students.
   7

2. Calculate the range for number of hours of sleep for the students.
   3.5

Students will use their knowledge of median and range and apply it to box plots throughout this lesson. Activate prior knowledge through intentional questioning:

- *What is the median of a data set? How can you arrange the data to determine the median?*
- *What does the median represent in this problem?*

10.1.2 Refresher: Box Plots

Component Overview: *Students make connections between box plots and the vocabulary used to describe them. In the previous unit, students used the vocabulary to interpret box plots. This component is an optional refresher, as the information was taught in the previous unit.*

10.1.2 Refresher: Box Plots

A blank box plot is shown. Label the blue boxes below using the words found in the bank.

median	first quartile (Q_1)	minimum (min)
third quartile (Q_3)	maximum (max)	

Consider having the students practice just finding the five-number summary before they must draw and label a box plot.

1. Complete the statements.

The *interquartile range* is a measure of spread determined by finding the distance between the first quartile and the third quartile. It measures the spread of the middle 50 *percent* of the data.

The *range* is another measure of spread determined by finding the difference between the largest data value and the smallest data value.

2. Complete each statement.

   a. 25% of the data falls between the minimum and the first quartile (Q_1).

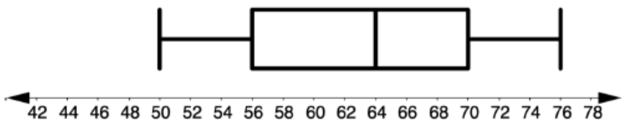
   b. 75% of the data falls between the minimum and the third quartile (Q_3).

   c. 50% of the data falls between the first quartile (Q_1) and the third quartile (Q_3).

10.1.2 Refresher: Box Plots (continued)

3. The box plot shows the distribution of the daily number of people who were at a large community pool in July 2018.

Number of Patrons at Community Pool



   a. What is the median number of people at the pool each day?
   64

   b. What is the maximum number of people?
   76

   c. What is the minimum number of people?
   50

   d. Determine the value of the first quartile (Q_1) from the box plot.
   56

   e. Determine the value of the third quartile (Q_3) from the box plot.
   70

The **five-number summary** are the values of the minimum, first quartile (Q_1), median, third quartile (Q_3), and the maximum.
These are the values used to create a box plot to represent a data set.

What does it mean if the median line is not in the middle of the box?

Have the students tell you in words how to find the first and third quartile.

Consider a foldable or type of interactive notebook for the five-number summary. Examples, non-examples, and verbal descriptions will provide multiple pathways for students to engage with the content in meaningful ways that encourage conceptual understanding over rote memorization.

10.1.3 Guided Instruction: The Five-Number Summary

Component Overview: *This component combines all the aspects of finding the five-number summary. During this guided instruction, students work with a partner to apply the definitions of the five-number summary to various data sets. Students then use this information to analyze and interpret the data.*

10.1.3 Guided Instruction: The Five-Number Summary

1. The ages of 20 customers who attended a special event at a local restaurant are shown in order from least to greatest.

$$7, 8, 9, 10\ 10, 11, 12, 15, 16, 20, 20,$$
$$22, 23, 24, 28, 30, 33, 35, 38, 42$$

   a. Find the five-number summary of the data set.

   Minimum: 7

   First quartile: 10.5

   Median: 20

   Third quartile: 29

   Maximum: 42

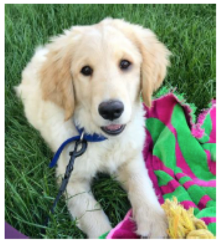
   b. Discuss with your partner if the data the restaurant collected is reflective of the ages of customers who eat at the restaurant.
   Answers will vary. It is reflective of the special event. It is not reflective of normal traffic at this restaurant.

Truthful data is data that is random, representative of the population, and has a large sample size.

This component provides auditory entry points for learners through discussion. Visual entry points can be provided by encouraging students to represent data using a box plot. Connecting the model to the vocabulary will help students synthesize the vocabulary rather than simply memorizing the definitions.

10.1.3 Guided Instruction: The Five-Number Summary (continued)

2. The birth weights, in ounces, of golden retriever puppies born at a kennel were recorded for the past month. The data is recorded in the table.



13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		

   a. Find the five-number summary of the data set.

      Minimum: 13

      First quartile: 16

      Median: 17

      Third quartile: 18

      Maximum: 20

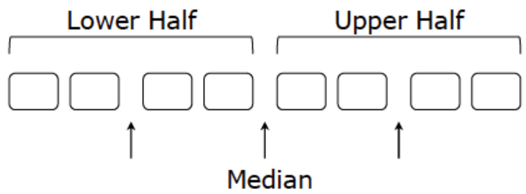
   b. Discuss with your partner why the data may not be a true representation of the birth weight of all puppies.
      Answers will vary. Because this is a single breed, Golden Retriever, it does not represent all breeds. The size of the parents may have an effect on the birth weights of the puppies.

3. What differences did you notice when calculating the five-number summary for the two different data sets?
   Answers will vary. The first data set has a larger range. The second data set has more entries.

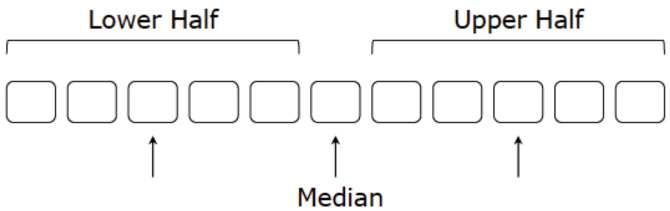
As students are working together to complete this component, encourage them to use mathematical vocabulary and language while explaining their process for finding the five-number summary with their partner. This component should help students formalize their understanding of the definitions as they continue to apply them to analyze data sets.

10.1.3 Guided Instruction: The Five-Number Summary (continued)

An illustration of the five-number summary for an even data set is shown.



An illustration of the five-number summary for an odd data set is shown.



4. Discuss with your partner your observations and any questions you have about the illustrations.
Answers will vary. The median and first and third quartiles will not always be equal to one of the values in the data set. The quartile may fall between two values, in which case the average is found to determine the value of the quartile.

Notice and Wonder
Consider structuring this component with a Notice and Wonder routine. Students take a few moments to observe each illustration and make notes of what they notice about it and what they wonder. Then they discuss with a partner and compare their responses. After students have had a chance to share with a partner, bring the class together and record notices and wonderings on chart paper or a white board and draw attention to themes in student responses.
Students commonly divide the data incorrectly when calculating quartile values, causing their constructed box plots to be incorrect.

10.1.4 Guided Practice: Creating a Box Plot

Component Overview: *In this Guided Practice, students follow steps to create a box plot using provided data. Use informal pieces of feedback or data by questioning students from prior components. Consider mobilizing some students to practice independently, some students to practice in pairs, and some students to be in a small group with the teacher.*

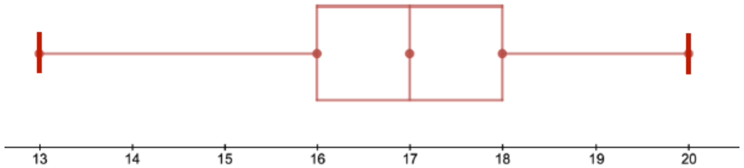
10.1.4 Guided Practice: Creating a Box Plot

1. Using the data and five-number summary on the birth weight, in ounces, of the golden retriever puppies born at a kennel in the last month, create a box plot from the steps listed.

13	14	15	15	16	16	16	16	17
17	17	17	17	17	17	18	18	18
18	18	18	18	18	19	20		

Step 1: Label the axis by starting at a number slightly less than the minimum. Increase by the same interval, making sure each interval is equally spaced, until you reach (or pass) the maximum.

Birth weight, in ounces, of golden retriever puppies



Step 2: Place a dot above the axis for each value of the five-number summary.

Step 3: Draw the box from Q_1 to Q_3 with a line inside the box to represent the median.

Step 4: Connect the minimum and the maximum to each side of the box with a line.

Step 5: Include a title and a label for the axis.

2. Compare your box plots with your partner. Make any revisions if necessary.

Consider differentiating this component by ability. Use informal data gathered earlier in the lesson to group students. Allow students who demonstrate understanding to work independently, students who may benefit from discussion to work in pairs and pull a small group of students who are struggling to guide them through this component verbally.

10.1.5 Try It: Creating a Box Plot

Component Overview: *Students find the five-number summary of a set of data and apply it to create a box plot. Students then answer questions based on the box plot and data. Consider allowing students to work in pairs to encourage mathematical discourse as they work through this component.*

10.1.5 Try It: Creating a Box Plot

1. The ages of the 21 restaurant customers who attended a special event at a local restaurant are shown.

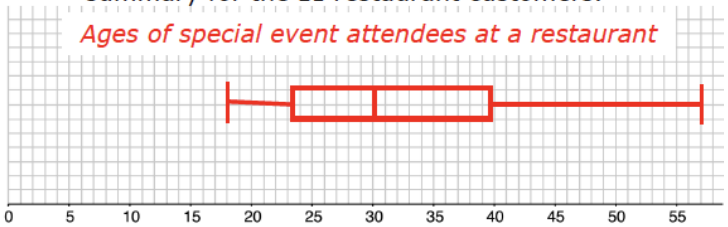
   27, 28, 39, 18, 18, 21, 32, 25, 46, 40, 40,
   22, 23, 24, 28, 30, 33, 35, 38, 42, 57

   a. Determine the five-number summary for the data set.

      18, 18, 21, 22, 23, 24, 25, 27, 28,28, 30,
      32, 33, 35,38,39,40, 40,42 46,57

      Minimum: 18
      First Quartile: 23.5
      Median: 30
      Third Quartile: 39.5
      Maximum: 57

   b. Create a box plot using the data and five-number summary for the 21 restaurant customers.



   c. Does the length of each part of the box plot need to be the same?
      No

   d. Why is the actual length of each part not equal, even though they each represent 25% of the data?
      Answers will vary. There are less numbers in the data set that are further from the median, causing the data to not be divided equally. The distribution of the ages of the customer will determine how the boxes are drawn. Ages closer together will skew the box one way or another.

   e. What is the interquartile range for the data?
      16

   f. What is the range?
      39

Students should first recognize that the data needs to be listed in ascending order to determine the five-number summary.

Circulate to ensure students are using the five-number summary to create their box plot. To deepen conceptual understanding, ask questions like:

- *How can you determine where the box plot whisker should start? How can you determine where it should end?*
- *Will the median be in the middle of the box plot? Why or why not?*
- *How can you determine where the quartiles should be?*

10.1.6 Your Turn!

Component Overview: *Students are given the opportunity to take what was practiced in the Guided Practice and Try It components and apply their knowledge as they work with a partner. Probe students as they work using assessing and critical thinking questions to check for understanding.*

10.1.6 Your Turn!

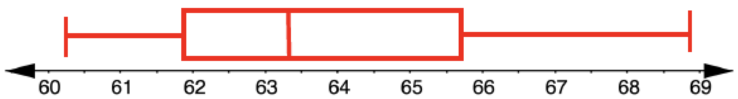
1. Tom Brady is considered by most to be one of the best NFL quarterbacks of all time. The five-number summary for his pass-completion percentages from 2000 to 2009 is shown.

Minimum	Maximum	Q_1	Q_3	Median
60.2	68.9	61.8	65.7	63.3

a. Discuss with your partner why the data used to determine the five-number summary for Tom Brady's pass completion is more truthful than the data for golden retriever puppies, or the ages of the restaurant customers who attended a special event.
Answers will vary. The data is more comprehensive, it is over a ten-year period. The puppies data consisted of one month of litters and the restaurant customers consisted of one special event.

b. Use the five-number summary to draw and label a box plot of the data set.

Tom Brady's Completion Percentages 2000-2009



c. Determine an alternative scale that could be used to make the box plot.
I could have also plotted by the whole number.

Within this benchmark, it is the expectation to use appropriate titles, labels, scales, and units when constructing graphical representations. Circulate to ensure students are using appropriate scales and making this connection.

10.1.6 Your Turn! (continued)

d. What scale would you use if the data values ranged from 10.3 to 48.4?
Answers will vary. By 2.

e. Label the number line with your scale from Part D. Then, compare it with your partner.

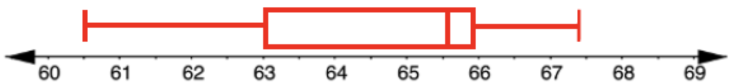
2. Tom Brady's pass-completion percentages from 2009 to 2019 are listed in the table.

Year	2009	2010	2011	2012	2013	2014
%	65.7	65.9	65.6	63.0	60.5	64.1
Year	2015	2016	2017	2018	2019	
%	64.4	67.4	66.3	65.8	60.8	

60.5, 60.8, 63.0, 64.1, 64.4, 65.6, 65.7, 65.8, 65.9, 66.3, 67.4
Minimum 60.5, *First Quartile* 63.0, *Median* 65.6, *Third Quartile* 65.9, *Maximum* 67.4.

a. Create a box plot of Tom Brady's pass-completion percentage.

Tom Brady's Pass-Completion Percentages 2009-2019



b. Determine the interquartile range for Tom Brady's pass-completion percentage.
2.9

c. What is the range of pass-completion percentages?
6.9

If you see students struggling to create the box plot, provide additional scaffolding through questioning:

- *What scale makes sense for these numbers? Why?*
- *How can you determine where the first quartile should start?*
- *How can you determine the median? Where will this be represented in the box plot?*

Consider a peer review process for students to check each other's work as an added layer of support.

10.1.7 Wrap-Up: Reflection

Component Overview: *Students reflect on their understanding of creating box plots. Students will need these skills when creating displays in the next several lessons.*

10.1.7 Wrap-Up: Reflection

Place an **X** on the line to indicate your understanding of creating box plots. Then provide a brief explanation or evidence to support your self-assessment.

1. I can find the five-number summary of a data set.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

Answers will vary.

2. I can create an axis with the appropriate scale.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

Answers will vary.

3. I can find and interpret the interquartile range.

I need help. I can't get started.	I'm getting there.	I understand and have evidence.

Answers will vary.

For students who might struggle representing their conceptual understanding through writing, consider alternative ways to assess or measure learning for this lesson. Students could choose to record themselves explaining the answer verbally, drawing and labeling, or providing an example. All forms can show an equal level of rigor and conceptual understanding.